



Jeremías Likerman

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Profile

Geoscientist (PhD, 2015) specializing in structural geology, geomechanics, and multiscale numerical modelling of the Andean orogen, based at CONICET-IDEAN (Argentina), an institution eligible under this call. Fifteen years of Andean field research address two of the six IGCP priority themes: **Geodynamics** (subduction dynamics, lithospheric deformation, and mantle-plume interactions along the Andean margin) and **Earth Resources** (natural-fracture prediction and connectivity in the Vaca Muerta unconventional play, and geothermal-potential assessment of volcanic systems). A sustained record of teaching, professional training, thesis supervision, and prior direct collaboration with researchers in Chile supports the regional-cooperation and capacity-building priorities of this call.

Education

PhD in Geological Sciences

UNIVERSITY OF BUENOS AIRES

Argentina

2010–2015

- Thesis: *Prediction of fracturing on irregular geological surfaces using static methods*. Supervisor: Dr. Ernesto Cristallini.

Postdoctoral Fellowship

CONICET AND UNIVERSITAT POLITÈCNICA DE CATALUNYA

Argentina/Spain

2015–2017

- Thermal evolution of oceanic lithosphere. Supervisors: Dr. Ernesto Cristallini and Dr. Sergio Zlotnik.

BSc in Geological Sciences

UNIVERSITY OF BUENOS AIRES

Argentina

2005–2010

- Thesis: *Geology and structure of the Tres Cruces compound anticline, Jujuy, Argentina*.

Current Appointments

Adjunct Researcher (Investigador Adjunto)

CONICET-IDEAN

Argentina

2018–present

- Fracture prediction, geodynamic modelling, and geothermal-potential modelling. Promoted to Investigador Adjunto, Nov. 2023.

Associate Professor

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

Argentina

May 2026–present

- Courses: Geostatistics and Geoinformatics.

Institutional Coordinator

IDEAN, CONICET-UBA

Argentina

Sep 2025–present

- Institute council member since April 2024 — direct experience in research-institute governance and administration.

Research Projects (Selected) — IGCP-Aligned

2026–28	Tectonic-structural analysis through analogue and numerical modelling: contributions to energy resources. PIP 11220250100817CO, CONICET	Argentina
2025	Geothermal potential of volcanic zones (PI) — <i>Earth Resources</i> . AECT-2025-1-0009	Spain
2025–26	Structural controls in volcanic systems: intrusions and damage zones linked to hydrocarbon generation (Co-PI) — <i>Earth Resources</i> . Williams Foundation	Argentina
2024–26	Thermomechanical assessment of geothermal potential in volcanic zones (PI) — <i>Earth Resources</i> . LINCGL2024	Spain
2024	Geodynamic modelling of subduction zones and mantle plumes (PI) — <i>Geodynamics</i> . AECT-2024-2-0027	Spain
2023	Geodynamics of the Adria microplate, from mantle to surface (Co-PI) — <i>Geodynamics</i> . GEO3BCN	Spain
2022	Geodynamic modelling of subduction zones (PI) — <i>Geodynamics</i> . AECT-2022-3-0023	Spain
2018	Natural fracturing and first-order structures, Cerro Domuyo (Co-PI) — <i>Earth Resources</i> . UBACyT	Argentina
2016	Numerical modelling of the thermal evolution of oceanic lithosphere (PI) — <i>Geodynamics</i> . PICT 2015-1226	Argentina

Five of the nine projects above were led as Principal Investigator, demonstrating a consistent track record of securing and managing competitive, internationally co-funded research grants.

Capacity-Building, Teaching & Regional Engagement

TEACHING AND PROFESSIONAL TRAINING

Geology of Unconventional Reservoirs

INSTRUCTOR IN CHARGE

Argentina
2025

- Professional training course on the geological characterization of shale reservoirs.

Advanced Research Field Trip

FIELD INSTRUCTOR

Argentina
2025

- Organizer and instructor, *Perspectives on Vaca Muerta: subsurface attributes analysed from surface exposures*.

Geostatistics, Geoinformatics, Geological Surveying, Introduction to Geology

DEPARTMENT OF GEOLOGICAL SCIENCES, UBA

Argentina
2013–present

- Continuous university teaching, from Teaching Assistant to Associate Professor.

SUPERVISION OF EARLY-CAREER RESEARCHERS Supervised 1 postdoctoral researcher (Plotek, B., 2024), 1 doctoral thesis (Correa Luna, C., 2023), and 11 undergraduate theses (2015–present, several ongoing). Currently mentoring 1 CONICET Assistant Researcher (Plotek, B., since Aug. 2025).

REGIONAL AND INTERNATIONAL COOPERATION

- Chile:** multi-year collaboration (2012–2017) with researchers at Universidad de Chile (L. Pinto, R. Charrier) on Andean tectonic inversion, resulting in a co-authored article in the *Journal of South American Earth Sciences* (2017) and a chapter in *Geological Society, London, Special Publications* v. 399 (2014).
- Participant, SZNet 2025 Chile Field Trip, NSF SZ4D initiative — ongoing engagement with the regional subduction-zone research network.
- Spain (UPC):** postdoctoral research and recurring visiting stays (2015, 2017, 2019, 2022) at the Numerical Computation Laboratory, Universitat Politècnica de Catalunya (BarcelonaTech), under Dr. Sergio Zlotnik — a decade-long bilateral cooperation.
- Coordinator, CONICET–CSIC (Spain) bi-national cooperation on geothermal modelling (2024–2026) — direct experience running an internationally co-funded, multi-year cooperation project of comparable scope to this call.
- Incoming Research Fellowship, Kobe University, Japan (JSPS, 2027) — a further, independently competitive international fellowship, reflecting a sustained track record of securing international research placements.

Publications

Complete list of 19 peer-reviewed journal articles (2013–2026), reverse chronological order.

- Villarroel, M.A., Browning, J., Jara, P.P., Harnett, C.E., Holohan, E.P., Marquardt, C.J., Guzman, M., & **Likerman, J.** (2026). The role of compressional structures in the development of collapse calderas: Insights from analogue models. *Journal of Volcanology and Geothermal Research*, 473, 108582.
- Gianni, G.M., Gallo, L.C., **Likerman, J.**, Echaurren, A., Gianni, C.R., & Faccena, C. (2025). Slab underthrusting is the primary control on flat-slab size. *Science Advances*, 11(28).
- Correa-Luna, C., Yagupsky, D.L., **Likerman, J.**, & Barcelona, H. (2025). Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play, Neuquén Basin, Argentina. *Tectonophysics*, 230640.
- Plotek, B., **Likerman, J.**, & Cristallini, E. (2024). Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model. *Journal of Structural Geology*, 105064.
- Navarrete, C., Gianni, G., Tassara, S., Zaffarana, C., **Likerman, J.**, Márquez, M., Wostbrock, J., Planavsky, N., Tardani, D., & Perez Frasette, M. (2024). Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province. *Earth-Science Reviews*, 249, 104651.
- Gianni, G.M., **Likerman, J.**, Navarrete, C.R., Gianni, C.R., & Zlotnik, S. (2023). Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction. *Nature Communications*, 14, 2083.
- Plotek, B., Heckenbach, E., Brune, S., Cristallini, E., & **Likerman, J.** (2022). Kinematics of fault-propagation folding: Analysis of velocity fields in numerical modeling simulations. *Journal of Structural Geology*, 162, 104703.
- Correa-Luna, C., Yagupsky, D.L., **Likerman, J.**, & Barcelona, H. (2022). Nat-

ural fracture characterization of the Los Catutos Member (Vaca Muerta Formation), Sierra de la Vaca Muerta, Neuquén Basin. *Journal of South American Earth Sciences*, 120, 104081.

9. **Likerman, J.**, Zlotnik, S., & Li, C.-F. (2021). The effects of small-scale convection in the shallow lithosphere of the North Atlantic. *Geophysical Journal International*, 227(3), 1512–1522.
10. García Morabito, E., Beltrán-Triviño, A., Terrizzano, C.M., Bechis, F., **Likerman, J.**, Von Quadt, A., & Ramos, V.A. (2021). The influence of climate on the dynamics of mountain building within the Northern Patagonian Andes. *Tectonics*, 40(2), e2020TC006374.
11. Luna, C.C., Yagupsky, D.L., & **Likerman, J.** (2019). Fracture network analysis of Yacoraite Formation in the Tres Cruces sub-basin, northwestern Argentina. *Journal of South American Earth Sciences*, 102446.
12. Jara, P., **Likerman, J.**, Charrier, R., Herrera, S., Pinto, L., Villarroel, M., & Winocur, D. (2017). Closure type effects on the structural pattern of an inverted extensional basin of variable width: analogue models. *Journal of South American Earth Sciences*.
13. Gianni, G.M., Echaurren, A., Folguera, A., **Likerman, J.**, Encinas, A., García, H.P.A., Dal Molin, C., & Valencia, V.A. (2017). Cenozoic intraplate tectonics in Central Patagonia: Record of main Andean phases in a weak upper plate. *Tectonophysics*, 721, 151–166.
14. Terrizzano, C.M., García Morabito, E., Christl, M., **Likerman, J.**, Tobal, J., Yamin, M., & Zech, R. (2017). Climatic and tectonic forcing on alluvial fans in the Southern Central Andes. *Quaternary Science Reviews*, 172, 131–141.
15. Cilona, A., Aydin, A., **Likerman, J.**, Parker, B., & Cherry, J. (2016). Structural and statistical characterization of joints and multi-scale faults in an alternating sandstone and shale turbidite sequence at the Santa Susana Field Laboratory. *Journal of Structural Geology*, 85, 95–114.
16. Tobal, J.E., Folguera, A., **Likerman, J.**, Naipauer, M., Sellés, D., Boedo, F.L., Ramos, V.A., & Gimenez, M. (2015). Middle to late Miocene extensional collapse of the North Patagonian Andes (41°30'–42° S). *Tectonophysics*, 657, 155–171.
17. Ghiglione, M.C., **Likerman, J.**, Barberón, V., Giambiagi, L.B., Aguirre-Urreta, B., & Suarez, F. (2014). Geodynamic context for the deposition of coarse-grained deep-water axial channel systems in the Patagonian Andes. *Basin Research*, 26(6), 726–745.
18. Jara, P., **Likerman, J.**, Winocur, D., Ghiglione, M.C., Cristallini, E.O., Pinto, L., & Charrier, R. (2014). Role of basin width variation in tectonic inversion: implications for the Abanico Basin, 32°–34° S, Central Andes. *Geological Society, London, Special Publications*, 399, SP399–7.
19. **Likerman, J.**, Burlando, J.F., Cristallini, E.O., & Ghiglione, M.C. (2013). Along-strike structural variations in the Southern Patagonian Andes: Insights from physical modeling. *Tectonophysics*, 590, 106–120.

Fellowships & Distinctions

2027	Research Fellowship in Japan Kobe University	JSPS
2025	SZNet 2025 Chile Field Trip National Science Foundation	SZ4D
2022	Academic Mobility Fellowship among Partner Institutions Ibero-American University Association for Postgraduate Studies	AUIP
2016	International Research Stay National Scientific and Technical Research Council	CONICET
2015	Postdoctoral Fellowship National Scientific and Technical Research Council	CONICET
2013	Doctoral Research Stay at Stanford University Fulbright – Bunge & Born Foundation	Doctoral fellowship

Skills & Languages

Modelling: Underworld2, LaMEM, GOLEM, ASPECT; analogue modelling, geomechanics, finite-element modelling; HPC workflows.

Spatial analysis & fieldwork: QGIS, GMT, PostGIS; seismic interpretation, geological mapping, UAV-assisted methods; 15+ years leading Andean field campaigns.

Programming: Python, R, MATLAB, $\LaTeX$, HTML5.

Languages: Spanish (native), English (fluent, C2), Italian (A2), Catalan (A2).

Alignment with Call Priorities

Priority themes: direct, ongoing research programme in Geodynamics (subduction dynamics, mantle-plume interactions, lithospheric deformation) and Earth Resources (natural-fracture prediction in unconventional reservoirs; geothermal-potential assessment), evidenced by a Principal-Investigator-led project track record across five internationally co-funded grants.

Capacity-building: continuous university teaching (2013–present), a dedicated professional-training course and field-based instruction for early-career geologists, and supervision of one postdoctoral researcher, one doctoral thesis, and eleven undergraduate theses.

Regional cooperation: documented, multi-year collaboration with Universidad de Chile researchers producing peer-reviewed outputs within the same publication series as a prior IGCP project (586-Y), continued engagement with the NSF SZ4D regional subduction-zone network, and current management of a bi-national (Argentina–Spain) research cooperation, providing direct experience relevant to leading a binational or regional IGCP project across Latin America and the Caribbean.